

STANDARD COLLEGE NTUNGAMO

S4 ACTIVITY OF INTEGRATION

INTRODUCTION TO ELECTRICITY

TIME : 40 minutes

At **Standard College Ntungamo**, Senior Four class wants to decorate their party room with small electric bulbs powered by **an accumulator**. However, they do not have enough connecting wires and they discovered that the source of emf is not charged and failed to supply current. The teacher asks the learners to investigate whether **other materials available around the school** can allow electric current to pass through them. The learners collect several materials such as **copper wire, an iron nail, aluminium foil, plastic ruler, rubber band, wooden stick, graphite from a pencil, and a piece of cardboard**.

Using the accumulator, **bulb, switch, and connecting wires**, they set up a simple electric circuit and insert each material in the circuit one at a time to see whether the **bulb lights**.

TASK

As a learner of physics:

1. Describe what must be done to restore the source of emf so that it can supply current again.
2. Draw a **simple circuit diagram** the learners could use for this investigation.
3. Explain **what happens in the circuit when the bulb lights**.
4. From the activity, classify the tested materials into:
 - (a) **Conductors**
 - (b) **Insulators**

END